

CODE

```
import tkinter as tk

from tkinter import ttk, messagebox

import os


waste_data = {
    "Plastic": [],
    "Paper": [],
    "Glass": [],
    "Metal": [],
    "Organic": []
}


recyclable = {"Plastic", "Paper", "Glass", "Metal"}

file_name = "waste_records.txt"


def add_record():
    name = name_entry.get().strip()
    category = category_box.get()
    quantity = quantity_entry.get().strip()
    date = date_entry.get().strip()

    if name == "" or quantity == "" or date == "":
        messagebox.showwarning("Warning", "Please enter all details.")
        return

    try:
        quantity = float(quantity)

        if quantity <= 0:
```

```
messagebox.showwarning(
    "Warning",
    "Quantity must be greater than 0."
)
return
```

```
except ValueError:
    messagebox.showerror(
        "Error",
        "Please enter a valid quantity."
    )
    return
```

```
record = (name, quantity, date)
waste_data[category].append(record)
```

```
messagebox.showinfo(
    "Success",
    "Waste record added successfully."
)
```

```
clear_boxes()
show_records()
```

```
def show_records():
    output.delete("1.0", tk.END)

    output.insert(
        tk.END,
        "===== WASTE COLLECTION RECORDS =====\n\n"
```

```
)
```

```
found = False
```

```
for category in waste_data:
```

```
    records = waste_data[category]
```

```
    if len(records) > 0:
```

```
        found = True
```

```
        output.insert(
```

```
            tk.END,
```

```
            category.upper() + "\n"
```

```
        )
```

```
    for record in records:
```

```
        name = record[0]
```

```
        quantity = record[1]
```

```
        date = record[2]
```

```
        output.insert(
```

```
            tk.END,
```

```
            "Name: " + name +
```

```
            " | Quantity: " + str(quantity) +
```

```
            " kg | Date: " + date + "\n"
```

```
        )
```

```
    output.insert(tk.END, "\n")
```

```
if found == False:
```

```
    output.insert(tk.END, "No records available.")
```

```

def search_record():
    search_name = name_entry.get().strip().lower()

    if search_name == "":
        messagebox.showwarning(
            "Warning",
            "Enter a resident name first."
        )
        return

    output.delete("1.0", tk.END)

    found = False

    output.insert(
        tk.END,
        "===== SEARCH RESULT =====\n\n"
    )

    for category in waste_data:
        for record in waste_data[category]:

            name = record[0]
            quantity = record[1]
            date = record[2]

            if search_name in name.lower():

                output.insert(

```

```
        tk.END,  
        "Name: " + name + "\n"  
    )  
  
    output.insert(  
        tk.END,  
        "Category: " + category + "\n"  
    )  
  
    output.insert(  
        tk.END,  
        "Quantity: " + str(quantity) + " kg\n"  
    )  
  
    output.insert(  
        tk.END,  
        "Date: " + date + "\n\n"  
    )
```

```
found = True
```

```
if found == False:
```

```
    output.insert(  
        tk.END,  
        "No record found."  
    )
```

```
def update_record():
```

```
    search_name = name_entry.get().strip().lower()  
    new_quantity = quantity_entry.get().strip()
```

```
if search_name == "" or new_quantity == "":
```

```
    messagebox.showwarning(
```

```
        "Warning",
```

```
        "Enter name and new quantity."
```

```
    )
```

```
    return
```

```
try:
```

```
    new_quantity = float(new_quantity)
```

```
    if new_quantity <= 0:
```

```
        messagebox.showwarning(
```

```
            "Warning",
```

```
            "Quantity must be greater than 0."
```

```
        )
```

```
    return
```

```
except ValueError:
```

```
    messagebox.showerror(
```

```
        "Error",
```

```
        "Enter a valid quantity."
```

```
    )
```

```
    return
```

```
for category in waste_data:
```

```
    records = waste_data[category]
```

```
    for i in range(len(records)):
```

```
        name = records[i][0]
```

```
        date = records[i][2]
```

```
if name.lower() == search_name:
```

```
    records[i] = (  
        name,  
        new_quantity,  
        date  
    )
```

```
    messagebox.showinfo(  
        "Success",  
        "Record updated successfully."  
    )
```

```
    show_records()  
    return
```

```
messagebox.showwarning(  
    "Not Found",  
    "Record not found."  
)
```

```
def classify_waste():
```

```
    category = category_box.get()
```

```
    if category in recyclable:
```

```
        messagebox.showinfo(  
            "Classification",  
            category + " is recyclable waste."  
        )
```

else:

```
    messagebox.showinfo(
        "Classification",
        category + " is organic waste."
    )
```

def recyclable_quantity():

```
    total = 0
```

```
    for category in recyclable:
```

```
        for record in waste_data[category]:
```

```
            total = total + record[1]
```

```
    output.delete("1.0", tk.END)
```

```
    output.insert(
```

```
        tk.END,
```

```
        "==== RECYCLABLE WASTE =====\n\n"
```

```
    )
```

```
    output.insert(
```

```
        tk.END,
```

```
        "Total recyclable waste: " +
```

```
        str(total) +
```

```
        " kg"
```

```
    )
```

def category_summary():

```
    output.delete("1.0", tk.END)
```



```
output.insert(  
    tk.END,  
    "===== CATEGORY SUMMARY =====\n\n"  
)
```

```
for category in waste_data:
```

```
    total = 0
```

```
    for record in waste_data[category]:
```

```
        total = total + record[1]
```

```
    output.insert(  
        tk.END,
```

```
        category + " : " +  
        str(total) +
```

```
        " kg\n"  
    )
```

```
def total_waste():
```

```
    total = 0
```

```
    for category in waste_data:
```

```
        for record in waste_data[category]:
```

```
            total = total + record[1]
```

```
    output.delete("1.0", tk.END)
```

```
    output.insert(  
        tk.END,
```

```
tk.END,  
"===== TOTAL WASTE =====\n\n"  
)
```

```
output.insert(  
tk.END,  
"Total waste collected: " +  
str(total) +  
" kg"  
)
```

```
def save_records():
```

```
file = open(file_name, "w")
```

```
for category in waste_data:
```

```
for record in waste_data[category]:
```

```
name = record[0]
```

```
quantity = record[1]
```

```
date = record[2]
```

```
file.write(  
category + "|" +
```

```
name + "|" +
```

```
str(quantity) + "|" +
```

```
date + "\n"
```

```
)
```

```
file.close()
```

```
messagebox.showinfo(  
    "Saved",  
    "Records saved successfully."  
)
```

```
def load_records():
```

```
    if os.path.exists(file_name) == False:  
        return
```

```
    file = open(file_name, "r")
```

```
    for line in file:
```

```
        data = line.strip().split(" | ")
```

```
        if len(data) == 4:
```

```
            category = data[0]
```

```
            name = data[1]
```

```
            try:
```

```
                quantity = float(data[2])
```

```
            except ValueError:
```

```
                continue
```

```
            date = data[3]
```

```
            if category in waste_data:
```

```
record = (  
    name,  
    quantity,  
    date  
)
```

```
waste_data[category].append(record)
```

```
file.close()
```

```
def clear_boxes():  
    name_entry.delete(0, tk.END)  
    quantity_entry.delete(0, tk.END)  
    date_entry.delete(0, tk.END)  
    category_box.set("Plastic")
```

```
def exit_program():  
    save_records()  
    window.destroy()
```

```
window = tk.Tk()  
window.title("Smart Waste Management")  
window.geometry("850x650")  
window.configure(bg="#e8f5e9")
```

```
title = tk.Label(  
    window,  
    text="SMART WASTE COLLECTION SYSTEM",
```

```
    font=("Arial", 20, "bold"),  
    bg="#2e7d32",  
    fg="white"  
)
```

```
title.pack(fill="x", pady=10)
```

```
input_frame = tk.Frame(  
    window,  
    bg="#e8f5e9"  
)
```

```
input_frame.pack(pady=10)
```

```
tk.Label(  
    input_frame,  
    text="Resident Name:",  
    font=("Arial", 12),  
    bg="#e8f5e9"  
)grid(row=0, column=0, padx=10, pady=8)
```

```
name_entry = tk.Entry(  
    input_frame,  
    width=30  
)
```

```
name_entry.grid(  
    row=0,  
    column=1,  
    padx=10,  
    pady=8
```

```
)
```

```
tk.Label(  
    input_frame,  
    text="Waste Category:",  
    font=("Arial", 12),  
    bg="#e8f5e9"  
).grid(row=1, column=0, padx=10, pady=8)
```

```
category_box = ttk.Combobox(  
    input_frame,  
    values=[  
        "Plastic",  
        "Paper",  
        "Glass",  
        "Metal",  
        "Organic"  
    ],  
    width=27,  
    state="readonly"  
)
```

```
category_box.set("Plastic")
```

```
category_box.grid(  
    row=1,  
    column=1,  
    padx=10,  
    pady=8  
)
```

```
tk.Label(  
    input_frame,  
    text="Quantity (kg):",  
    font=("Arial", 12),  
    bg="#e8f5e9"  
)grid(row=2, column=0, padx=10, pady=8)
```

```
quantity_entry = tk.Entry(  
    input_frame,  
    width=30  
)
```

```
quantity_entry.grid(  
    row=2,  
    column=1,  
    padx=10,  
    pady=8  
)
```

```
tk.Label(  
    input_frame,  
    text="Collection Date:",  
    font=("Arial", 12),  
    bg="#e8f5e9"  
)grid(row=3, column=0, padx=10, pady=8)
```

```
date_entry = tk.Entry(  
    input_frame,  
    width=30  
)
```

```
date_entry.grid(  
    row=3,  
    column=1,  
    padx=10,  
    pady=8  
)
```

```
button_frame = tk.Frame(  
    window,  
    bg="#e8f5e9"  
)
```

```
button_frame.pack(pady=10)
```

```
tk.Button(  
    button_frame,  
    text="Add Record",  
    width=18,  
    command=add_record,  
    bg="#4CAF50",  
    fg="white"  
)
```

```
tk.Button(  
    button_frame,  
    text="Update Record",  
    width=18,  
    command=update_record,  
    bg="#FF9800",  
    fg="white"  
)
```



```
tk.Button(  
    button_frame,  
    text="Search Record",  
    width=18,  
    command=search_record,  
    bg="#2196F3",  
    fg="white"  
)grid(row=0, column=2, padx=5, pady=5)
```

```
tk.Button(  
    button_frame,  
    text="Display Records",  
    width=18,  
    command=show_records,  
    bg="#607D8B",  
    fg="white"  
)grid(row=1, column=0, padx=5, pady=5)
```

```
tk.Button(  
    button_frame,  
    text="Classify Waste",  
    width=18,  
    command=classify_waste,  
    bg="#795548",  
    fg="white"  
)grid(row=1, column=1, padx=5, pady=5)
```

```
tk.Button(  
    button_frame,  
    text="Recyclable Quantity",
```

```
width=18,  
command=recyclable_quantity,  
bg="#009688",  
fg="white"  
)grid(row=1, column=2, padx=5, pady=5)
```

```
tk.Button(  
    button_frame,  
    text="Category Summary",  
    width=18,  
    command=category_summary,  
    bg="#673AB7",  
    fg="white"  
)grid(row=2, column=0, padx=5, pady=5)
```

```
tk.Button(  
    button_frame,  
    text="Total Waste",  
    width=18,  
    command=total_waste,  
    bg="#3F51B5",  
    fg="white"  
)grid(row=2, column=1, padx=5, pady=5)
```

```
tk.Button(  
    button_frame,  
    text="Save Records",  
    width=18,  
    command=save_records,  
    bg="#008000",  
    fg="white"
```

```
).grid(row=2, column=2, padx=5, pady=5)
```

```
output = tk.Text(  
    window,  
    width=90,  
    height=14,  
    font=("Courier New", 10)  
)
```

```
output.pack(  
    padx=20,  
    pady=10  
)
```

```
tk.Button(  
    window,  
    text="EXIT",  
    width=20,  
    command=exit_program,  
    bg="#D32F2F",  
    fg="white",  
    font=("Arial", 11, "bold")  
).pack(pady=5)
```

```
load_records()
```

```
window.mainloop()
```

OUTPUTS

Smart Waste Management

SMART WASTE COLLECTION SYSTEM

Resident Name:

Waste Category:

Plastic

Quantity (kg):

Collection Date:

Add Record

Update Record

Search Record

Display Records

Classify Waste

Recyclable Quantity

Category Summary

Total Waste

Save Records

===== WASTE COLLECTION RECORDS =====

PLASTIC
Name: skpr | Quantity: 10.0 kg | Date: 1.1.2026
METAL
Name: skp | Quantity: 52.0 kg | Date: 1.9.2026

EXIT

SMART WASTE COLLECTION SYSTEM

Resident Name:

Waste Category:

Glass

Quantity (kg):

Collection Date:

Add Record

Update Record

Search Record

Display Records

Classify Waste

Recyclable Quantity

Category Summary

Total Waste

Save Records

===== RECYCLABLE WASTE =====

Total recyclable waste: 122.0 kg

EXIT

SMART WASTE COLLECTION SYSTEM

Resident Name:

Waste Category:

Quantity (kg):

Collection Date:

Add Record

Update Record

Search Record

Display Record

Save

Recyclable Quantity

Category Summary

Save Records



Records saved successfully.

OK

===== TOTAL WASTE =====
Total waste collected: 122.0 kg

EXIT

SMART WASTE COLLECTION SYSTEM

Resident Name:

Waste Category:

Quantity (kg):

Collection Date:

Add Record

Update Record

Search Record

Display Records

Classify Waste

Recyclable Quantity

Category Summary

Total Waste

Save Records

===== CATEGORY SUMMARY =====

Plastic : 40.0 kg
Paper : 0 kg
Glass : 30.0 kg
Metal : 52.0 kg
Organic : 0 kg

EXIT

SMART WASTE COLLECTION SYSTEM

Resident Name:

Waste Category:

Quantity (kg):

Collection Date:

===== RECYCLABLE WASTE =====

Total recyclable waste: 122.0 kg

SMART WASTE COLLECTION SYSTEM

Resident Name:

Waste Category:

Quantity (kg):

Collection Date:

Classification

 Glass is recyclable waste.

===== WASTE COLLECTION RECORD =====

PLASTIC

Name: skpr | Quantity: 20.0 kg | Date: 2.2.2026

Name: skpr | Quantity: 20.0 kg | Date: 2.2.2026

GLASS

Name: skpr | Quantity: 30.0 kg | Date: 1.9.2026

METAL

Name: skp | Quantity: 52.0 kg | Date: 1.9.2026

SMART WASTE COLLECTION SYSTEM

Resident Name:

Waste Category:

Quantity (kg):

Collection Date:

Add Record

Update Record

Search Record

Display Records

Classify Waste

Recyclable Quantity

Category Summary

Total Waste

Save Records

===== WASTE COLLECTION RECORDS =====

PLASTIC

Name: skpr | Quantity: 20.0 kg | Date: 1.1.2026

Name: skpr | Quantity: 20.0 kg | Date: 2.2.2026

GLASS

Name: skpr | Quantity: 30.0 kg | Date: 1.9.2026

METAL

Name: skp | Quantity: 52.0 kg | Date: 1.9.2026

EXIT

SMART WASTE COLLECTION SYSTEM

Resident Name:

Waste Category:

Quantity (kg):

Collection Date:

Add Record

Update Record

Search Record

Display Records

Classify Waste

Recyclable Quantity

Category Summary

Total Waste

Save Records

===== SEARCH RESULT =====

Name: skpr

Category: Plastic

Quantity: 20.0 kg

Date: 1.1.2026

Name: skpr

Category: Plastic

Quantity: 20.0 kg

Date: 2.2.2026

Name: skpr

Category: Glass

EXIT

SMART WASTE COLLECTION SYSTEM

Resident Name:

Waste Category:

Quantity (kg):

Collection Date:

Add Record

Update Record

Search Record

Display Records

Classify Waste

Recyclable Quantity

Category Summary

Total Waste

Save Records

===== WASTE COLLECTION RECORDS =====

PLASTIC

Name: skpr | Quantity: 20.0 kg | Date: 1.1.2026

Name: skpr | Quantity: 20.0 kg | Date: 2.2.2026

GLASS

Name: skpr | Quantity: 30.0 kg | Date: 1.9.2026

METAL

Name: skp | Quantity: 52.0 kg | Date: 1.9.2026

EXIT